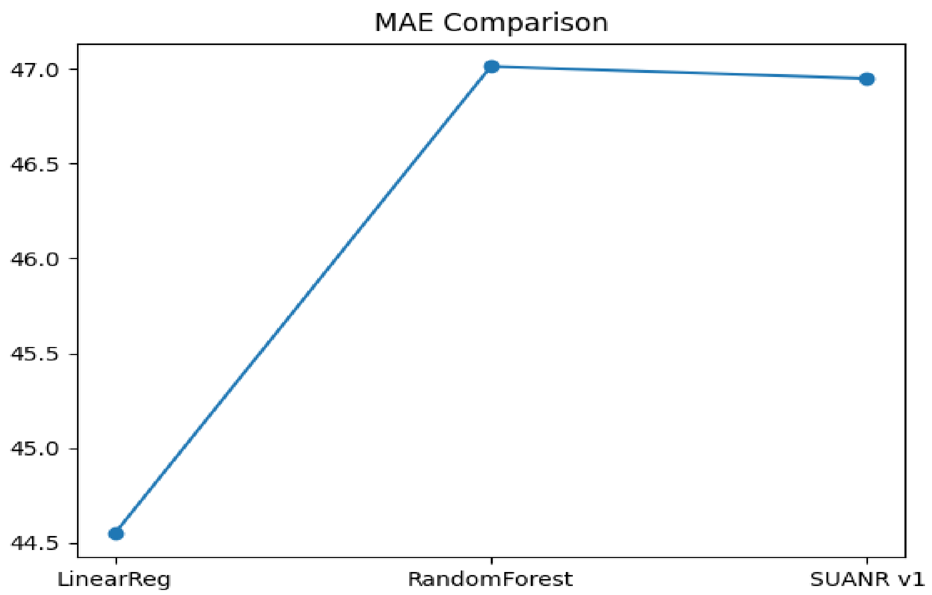
SUANR v1 Publication Package

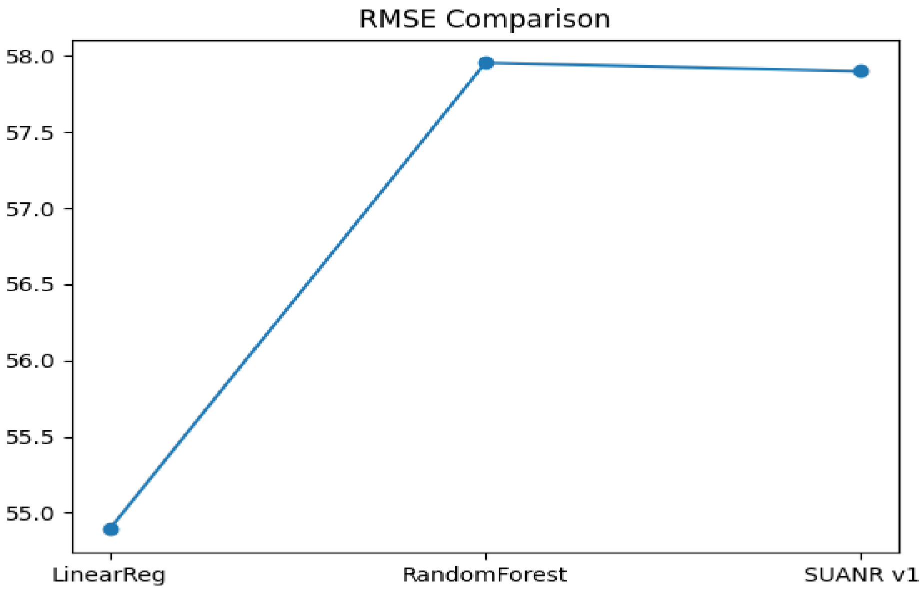
This document presents SUANR v1 (Stochastic Uncertainty-Aware Neural Regression), a probabilistic regression framework built under the MPRD (Multi-Path Recursive Dynamics) architecture with EDUM (Entropy-Driven Uncertainty Modulation). SUANR v1 is benchmarked against Linear Regression and Random Forest on a regression dataset.

Methodology: - MPRD Framework: multi-path representation and recursive dynamics for structured learning. - SUANR Algorithm: probabilistic regression model producing prediction + uncertainty estimates. - EDUM Mechanism: entropy-driven modulation of uncertainty signals.

Model	MAE	RMSE
Linear Regression	44.5463	54.8911
Random Forest	47.0133	57.9552
SUANR v1	46.9497	57.9000

Performance Figures




RMSE Comparison
58.0
57.5
57.0
56.5
56.0
55.5
55.0
LinearReg
RandomForest
SUANR v1

Conclusion: SUANR v1 demonstrates competitive performance against baseline models, closely matching Random Forest and approaching Linear Regression performance. This establishes a validated baseline for future development of SUANR v2 with adaptive uncertainty modeling via EDUM refinement.